AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

T216A
Cisco Networking (CCCNA) part A

Final Examination
First Semester 2025/2026

Date: 19/Jan./2026

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 3h

Instructions:

- Total Marks: **50**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

Arab Open University

CamScanner

Part 1: Multiple choices. 1 mark per question. **Total: 5 Marks**

1. The purpose of the network security accounting function is _____
   a. To determine which resources a user can access
   b. To keep track of the actions of a user
   c. To provide challenge and response questions
   d. To require users to prove who they are
2. What important information is added to the TCP/IP transport layer header to ensure communication and connectivity with a remote network device?
   a. Destination and source logical network addresses
   b. Destination and source physical addresses
   c. Destination and source port numbers
   d. Timing and synchronization
3. What benefit does SSH offer over Telnet for remotely managing a router?
   a. Encryption
   b. Connections via multiple VTY lines
   c. Authorization
   d. TCP Usage
4. A network contains multiple VLANs spanning multiple switches. What happens when a device in VLAN 20 sends a broadcast Ethernet frame?
   a. Only devices in VLAN 20 see the frame
   b. All devices in all VLANs see the frame
   c. Devices in VLAN 20 and the management VLAN see the frame
   d. Only devices that are connected to the local switch see the frame
5. How do switch buffer affect network performance?
   a. They hold data temporarily when a collision occurs until normal data transmission resumes
   b. They store frames received, thus preventing premature frame discarding when congestion occurs
   c. They provide extra memory for a particular port if autonegotiation of speed or duplex fails
   d. They provide error checking on the data received

Part 2: Answer all the questions. 4 marks per question **Total: 16 Marks**

1) When a client boots (wants to join to a network), it begins a four-step process to obtain a lease. For this, the following fours messages are used:
   - DHCPDISCOVER
   - DHCPREQUEST
   - DHCPACK
   - DHCPOFFER

   a) Place these messages in the correct order when used to obtain a lease. (2 marks)
   b) Which of these four messages are sent as unicast during the lease process. (2 marks)
2) Answer the following questions related to the STP protocol :
   a) What is the purpose of the STP protocol? (1 mark)
   b) What is the name of the messages being exchanges by STP protocol? (2 marks)
   c) Based on which condition is the root bridge elected. (2 marks)
3) Answer the following questions:
   a) Which device stops the propagation of broadcast frames? (1 mark)

CamScanner

b) Explain how a switch decides to which port(s) a frame should be forwarded. (2 marks)

c) Which one of the three forwarding methods (cut-through, fragment free and store and forward) will check the destination and ensure that the frame is at least 64 Bytes? (1 mark)

4) Answer the following questions:

a) List two features of the TCP protocol (2 marks)

b) What is the use of the ICMP echo message? (2 marks)

Part 3: Answer all the following questions **Total: 29 Marks**

1. Answer the following questions related to VLSM (12 marks)

The following network topology was provided by the network administrator:

- **LAN1:** has 40 smart devices, 4 wireless servers connected via 4 access points*.
- **LAN2:** has 23 PCs, 23 VoIP, 4 servers, and 3 network printers connected to 2 switches*
- **LAN3:** has 8 servers connected to 1 switch*.
- **LAN4:** has 10 PCs, 10 VoIP phones, 10 network printers, 1 server connected to 1 switch*.

* Switch(es) and Access point(s) are accessible remotely (an IP address is needed)

a) Determine the number of hosts for each of the 4 networks? (4 marks) ***(Show the steps you followed to obtain the number of hosts)***

b) Determine the prefix length for each network (4 marks) ***(Show the steps you followed to obtain the prefix.)***

c) Subnet the address block 180.62.3.0/24 as appropriate to provide an efficient use of the addresses but at the same time provide sufficient addresses for all groups and links. Present your answer in a table similar to table1. (4 marks)

Network name	Network address	Host range	Broadcast address	Prefix (or subnet mask)

Table 1

2. Answer the following questions (8 marks)

a) Refer to figure A. Explain why, after configuring the EtherChannel on both switches, the EtherChannel was not successfully established. (4 marks)

Switch S1

```
S1(config)#interface range Fa0/1-3
S1(config-if-range)#channel-group 3 mode passive
S1(config-if-range)#exit
S1(config)#interface port-channel 3
S1(config-if)#switchport nonegotiate
```

Switch S2

```
S2(config)#interface range fa0/1-3
S2(config-if-range)#channel-group 2 mode active
S2(config)#interface port-channel 2
S2(config-if)#switchport mode trunk
S2(config-if)#switchport nonegotiate
```

Figure A

CamScanner

b) Explain how a switch decides to which port(s) a frame should be forwarded. (2 marks)

c) Which one of the three forwarding methods (cut-through, fragment free and store and forward) will check the destination and ensure that the frame is at least 64 Bytes? (1 mark)

4) Answer the following questions:

a) List two features of the TCP protocol (2 marks)

b) What is the use of the ICMP echo message? (2 marks)

Part 3: Answer all the following questions **Total: 29 Marks**

1. Answer the following questions related to VLSM (12 marks)

The following network topology was provided by the network administrator:

- **LAN1:** has 40 smart devices, 4 wireless servers connected via 4 access points*.
- **LAN2:** has 23 PCs, 23 VoIP, 4 servers, and 3 network printers connected to 2 switches*
- **LAN3:** has 8 servers connected to 1 switch*.
- **LAN4:** has 10 PCs, 10 VoIP phones, 10 network printers, 1 server connected to 1 switch*.

* Switch(es) and Access point(s) are accessible remotely (an IP address is needed)

a) Determine the number of hosts for each of the 4 networks? (4 marks)
(Show the steps you followed to obtain the number of hosts)

b) Determine the prefix length for each network (4 marks)
(Show the steps you followed to obtain the prefix.)

c) Subnet the address block 180.62.3.0/24 as appropriate to provide an efficient use of the addresses but at the same time provide sufficient addresses for all groups and links. Present your answer in a table similar to table1. (4 marks)

Network name	Network address	Host range	Broadcast address	Prefix (or subnet mask)

Table 1

2. Answer the following questions (8 marks)

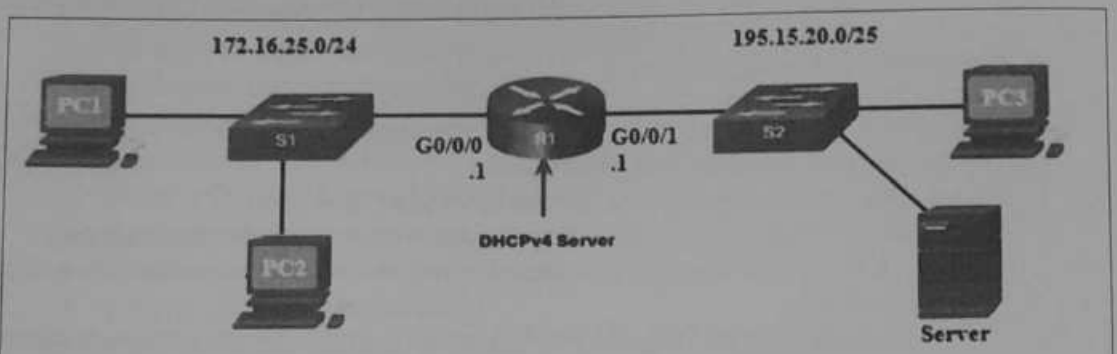
a) Refer to figure A. Explain why, after configuring the EtherChannel on both switches, the EtherChannel was not successfully established. (4 marks)

Switch S1
```
S1(config)#interface range Fa0/1-3
S1(config-if-range)#channel-group 3 mode passive
S1(config-if-range)#exit
S1(config)#interface port-channel 3
S1(config-if)#switchport nonegotiate
```
Switch S2
```
S2(config)#interface range fa0/1-3
S2(config-if-range)#channel-group 2 mode active
S2(config)#interface port-channel 2
S2(config-if)#switchport mode trunk
S2(config-if)#switchport nonegotiate
```

Figure A

CamScanner

b) Refer to figure B. R1 was configured as a DHCP server; explain why PC1 and PC2 are not able to ping any host from network 195.15.20.0/25 while they are able to successfully ping each. (4 marks)




```
Router#show running-config | section dhcp
ip dhcp excluded-address 172.16.25.1
ip dhcp excluded-address 172.16.25.100 172.16.25.254
ip dhcp pool Student
network 172.16.25.0 255.255.255.0
default-router 172.16.25.100
dns-server 172.16.25.100
domain-name student.com
Router#
```

Figure B

3. Answer the following questions (10 marks)
   a) Using the EUI-64 process and the MAC address A123:4567:D3E1 determine the interface ID of the IPv6 address. Show clearly the steps of calculation. (3 marks)
   b) From the interface ID calculated in part a determine the IPv6 link local address (1 marks)
   c) Write the command needed to configure a VLAN on a switch and to assign 3 interfaces to this VLAN. (4 marks)
      a. The VLAN number is 15
      b. The name of this VLAN is **advertisement**
      c. The interfaces assigned to this VLAN are: fa0/1, fa0/2 and fa0/3
   d) Complete the beneath configuration needed to configure **Router on a stick interVLAN Routing** for VLAN number 117 on interface G0/1/0. The sub-interface has the following IP address 192.168.1.1 and subnet mask 255.255.255.192 (2 marks)
      A. Interface _____
      B. _____ dot1Q 117
      C. Ip address _____
      D. Interface G0/1/0
      E. _____

End of questions

CamScanner